Technology

Reference: Intermediate Micro by Varian

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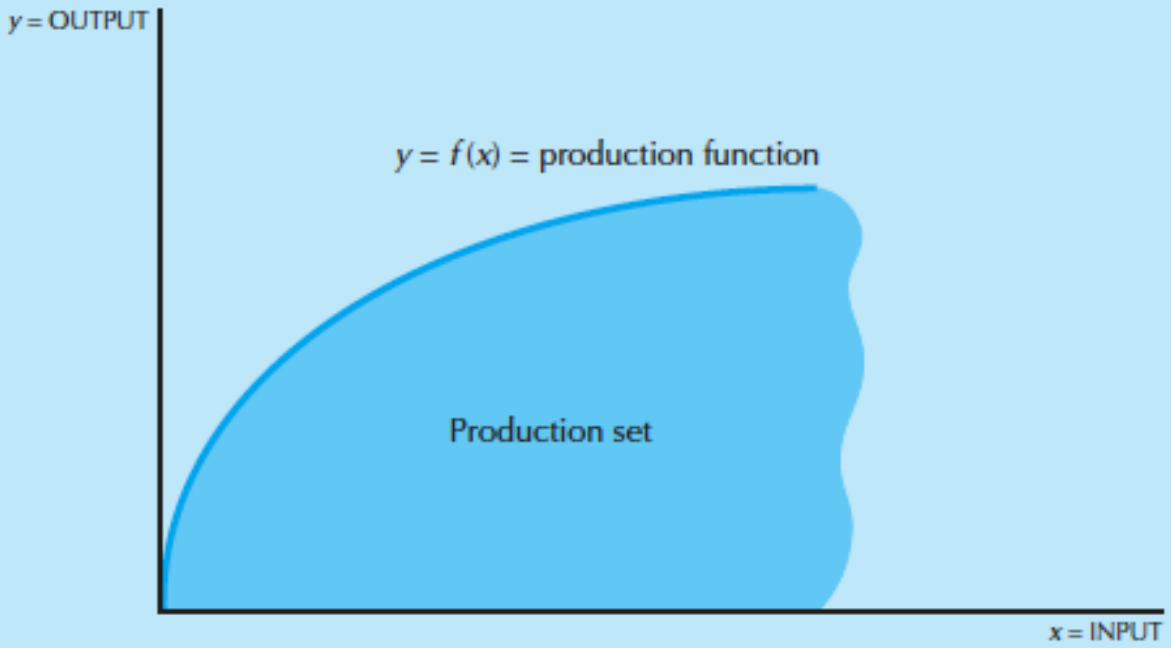
Inputs and Outputs

- Factors of production: Inputs.
- Labor and Capital.
- Physical vs financial capital.
- Typically flow units, but not needed right now.

Technological Constraints

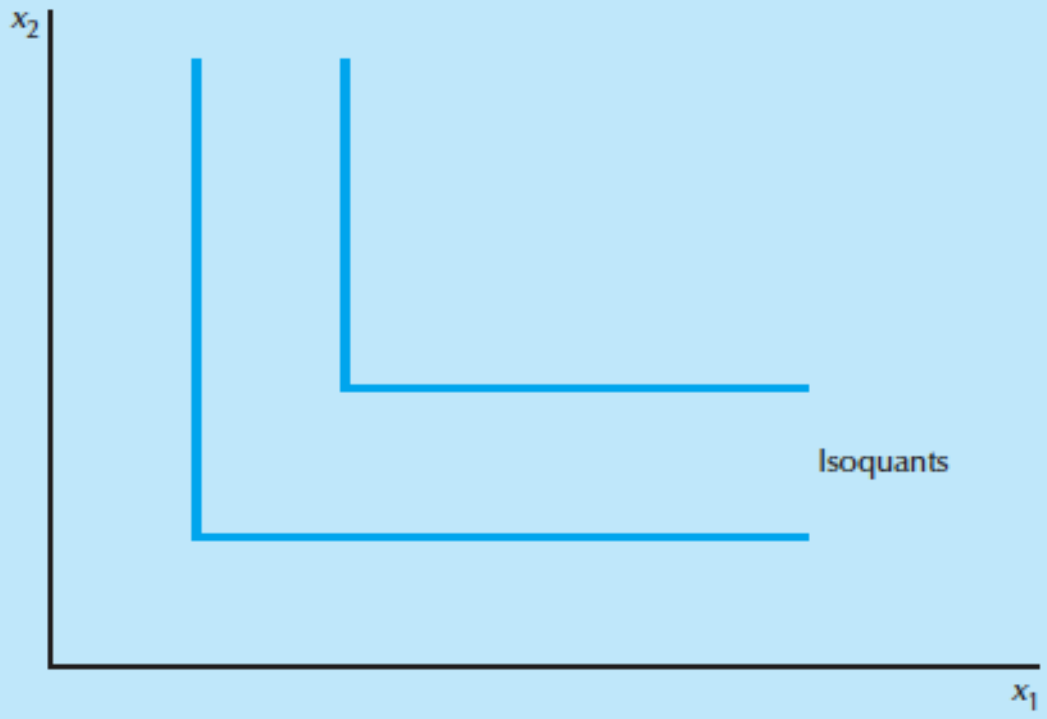
- Only certain combinations of inputs are feasible.
- All combinations of possible inputs are a production set. Naturally, they map to outputs.
- If there is one input x and one output y, then (x, y) represents the output for a given quantity of input.
- For more inputs, we have something like $y = f(x_1, x_2)$, where y is the maximum possible output for that combination of inputs.
- THis is the production function.

Production Set with Boundary



A production set. Here is a possible shape for a production set.

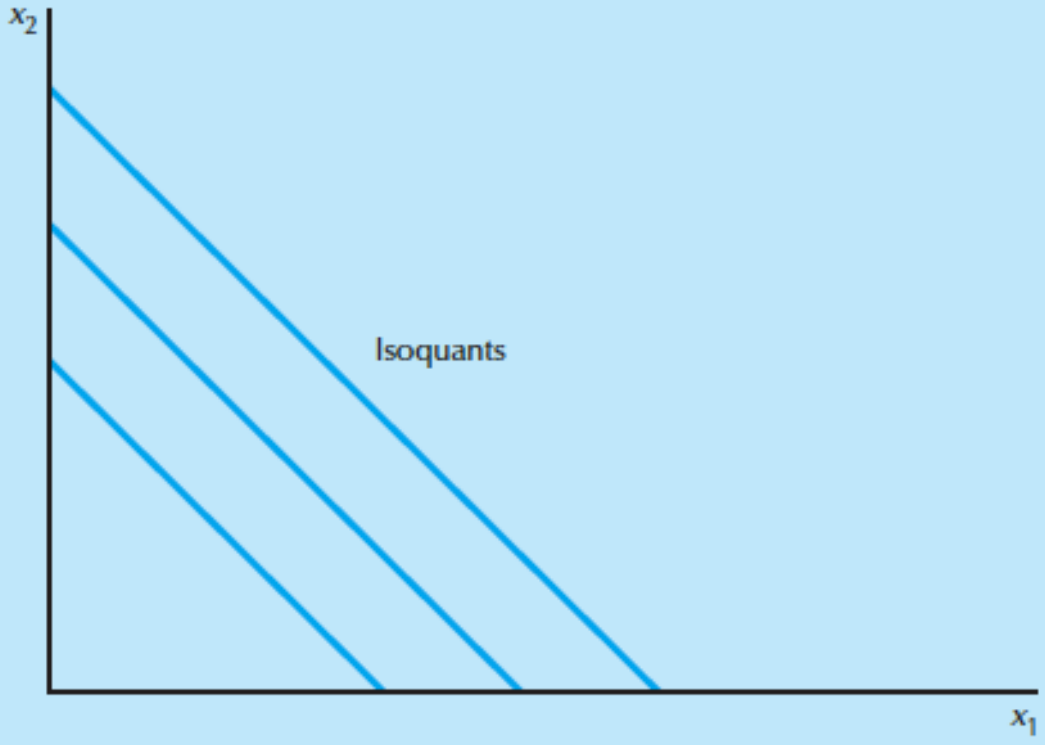
Fixed Proportions



Fixed proportions. Isoquants for the case of fixed proportions.

The isoquants look like those depicted in Figure 18.2. Note that these isoquants are just like the case of perfect complements in consumer theory.

Perfect Substitutes



Perfect substitutes. Isoquants for the case of perfect substitutes.

Cobb Douglas and Returns to Scale

$$f(x_1, x_2) = Ax_1^{\alpha} x_2^{\beta}$$

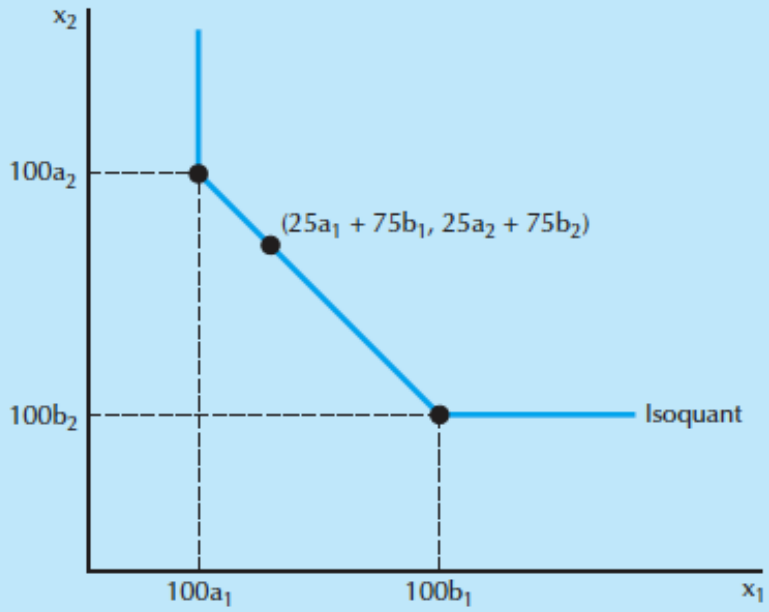
$$\alpha + \beta$$

is significant here.

- Some interesting properties to be explored.
- Can be generalized.

Properties of Technology

- Monotonic (weakly)
- Convex.



Convexity. If you can operate production activities independently, then weighted averages of production plans will also be feasible. Thus the isoquants will have a convex shape.

Marginal Product

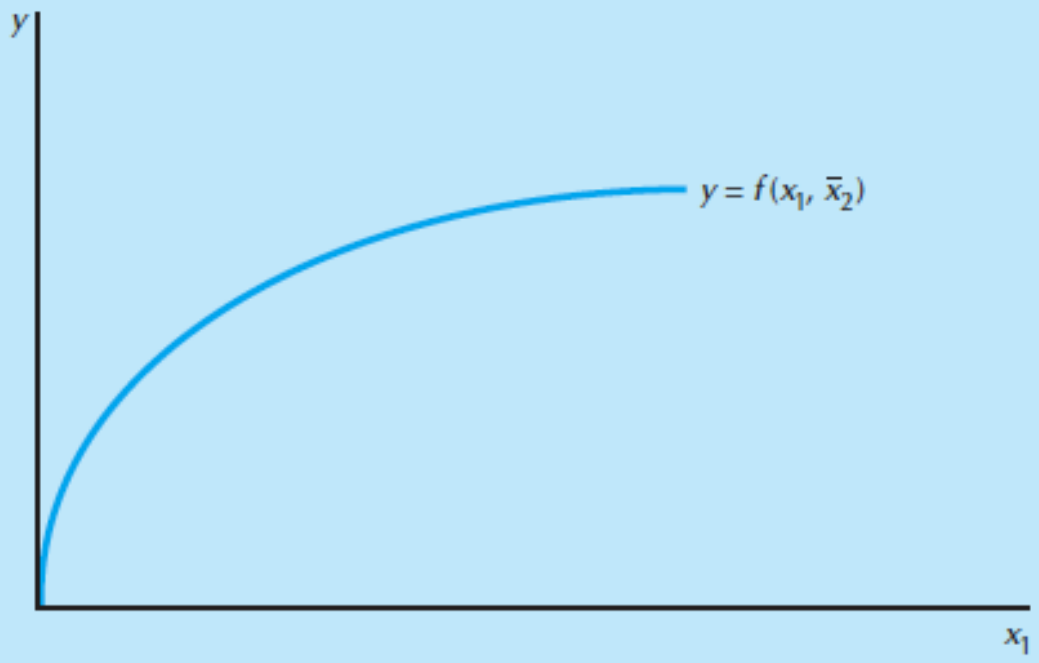
$$\frac{\Delta y}{\Delta x_1} = \frac{f(x_1 + \Delta x_1, x_2) - f(x_1, x_2)}{\Delta x_1}$$

- $MP_1(x_1, x_2)$.
- If we want to know the rate of substitution to keep production unaffected, we need

$$\Delta y = MP_1(x_1, x_2)\Delta x_1 + MP_2(x_1, x_2)\Delta x_2$$

$$TRS(x_1, x_2) = \frac{\Delta x_2}{\Delta x_1} = -\frac{MP_1(x_1, x_2)}{MP_2(x_1, x_2)}$$

When one input (Often Land or Large Capital) is fixed in the short run



Production function. This is a possible shape for a short-run production function.

SOme interesting equations

- $\pi = \sum_{i=1}^{n} p_i y_i - \sum_{i=1}^{n} w_i x_i$.
- Short Run: $\max_{x_1} \; pf(x_1, \bar{x}_2) - w_1 x_1 - w_2 \bar{x}_2$
- Long Run or no fixed input: $\max_{x_1, x_2} \; pf(x_1, x_2) - w_1 x_1 - w_2 x_2$